

Khalifeh Basiri

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EDUCATION

Carleton University

Bachelor of Computer Science, GPA: 3.5/4.0

Ottawa, ON
2026

WORK EXPERIENCE

Department of Fisheries and Oceans Canada

Software Developer Intern

Ottawa, ON

Sep. 2023 – Dec. 2025

- Engineered a multithreaded Python data explorer indexing and exporting metadata across 420 TB of Microsoft Azure Storage using Azure Blob Inventory, MongoDB, and SQLite; preserved UI responsiveness and the last valid database during failed refreshes.
- Reduced geographic-name validation time by 80% by building a bilingual, multithreaded Python application integrating the GNBC REST API and geospatial matching.
- Led a 3-person Agile team through the design, implementation, and integration of a user-requested Name Manager module; collaborated with GNBC stakeholders to improve API usability.
- Performed defect reproduction, root-cause analysis, regression testing, and user-acceptance testing (UAT) across multithreading, REST API, database, and geospatial workflows; validated staged releases with users across DFO's 5 regions and the Canadian Coast Guard.

Floxy.io

Freelance Frontend Developer

Remote

Apr. 2025 – May 2025

- Built and shipped a responsive Next.js marketing site supporting 20,000+ customers and reaching 23.9K monthly visits within four months of launch.
- Implemented mobile navigation, a proxy-type selector, auto-scrolling testimonials, language-specific API examples with copy-to-clipboard, reusable components, and centralized content models.

PROJECTS

ZDash | TypeScript, React, Electron, SQLite, OpenAI API, Zod

[Product Site](#)

- Engineered a Windows desktop diagnostics platform for Nissan and Infiniti CONSULT-II ECUs, implementing 9600-baud serial handshakes, framed-stream decoding for up to 22 live and derived channels, guarded fault-code workflows, and SQLite/WAL session recording.
- Integrated the OpenAI Responses API with complete vehicle context and Zod-validated, human-approved tools; developing grounded Retrieval-Augmented Generation (RAG) and evaluation workflows for automotive service documentation.

Real-Time Sign Language Translator | Python, TensorFlow, MediaPipe, Flask, React

[GitHub](#) | [Live Demo](#)

- Built a real-time ASL translation system that converts 21 hand landmarks into 63-feature prediction vectors using MediaPipe and TensorFlow Lite, serving confidence-scored inference through a Flask API and a live React webcam interface.

Collaborative Board | Next.js, TypeScript, Convex, Clerk, Tailwind CSS, Bun

[GitHub](#) | [Live Demo](#)

- Built and deployed a real-time community platform with Next.js, Convex, and Clerk, implementing reactive updates, server-side authorization, threaded comments, per-user vote state, ownership checks, and cascade deletion.

Tandem t:slim X2 Insulin Pump Simulator | C++, Qt, UML, Git

[GitHub](#)

- Engineered bolus and automated insulin-delivery logic within a 4-person C++/Qt team, implementing carbohydrate and correction calculations, manual and extended dosing, CGM-driven basal adjustments, safety limits, suspend/resume behavior, and alerts for out-of-range glucose and insulin overdose.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, C, SQL, HTML, CSS

Frameworks/Libraries: React, Next.js, Electron, Node.js, Flask, Qt, TensorFlow, TensorFlow Lite, MediaPipe, Tailwind CSS

Cloud/Developer Tools: Microsoft Azure, Azure Blob Storage, AWS, Docker, Git, Linux, Vercel, REST APIs, OpenAI API

Databases: MongoDB, SQLite, Oracle, Convex